



LifeGrid

# User Manual

How to use LifeGrid to share blood stock across a regional network, request blood for a patient, and give blood when a hospital nearby needs your group.

|         |                                               |
|---------|-----------------------------------------------|
| Version | 1.0 · phases 0–6                              |
| Date    | 3 October 2026                                |
| Covers  | Staff web app · donor and requester phone app |

Every screen in this manual comes from the demo build. All sites, units, donors, requests and phone numbers are synthetic. LifeGrid supports decisions; it does not replace clinical judgment, a licensed blood bank system, or local regulations.

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# About LifeGrid

Blood is perishable and type-specific. When every hospital manages it alone, one site throws away expiring units while another runs short. LifeGrid gives the whole region one live view of stock and helps people move it, request it and donate it in time.

## What you can do with it

- **See the network.** Stock at every hospital and blood bank, by blood group and by how soon it expires.
- **Plan ahead.** A 7-day demand forecast for every hospital, and a nightly transfer plan that moves units before they expire or before a hospital runs short. A manager approves every move.
- **Request blood.** A family member or a hospital asks for blood for a patient. The hospital confirms, LifeGrid checks stock across the network first, and only asks donors for what stock can't cover.
- **Ask donors when stock runs low.** Staff send an appeal; eligible donors nearby get a text message and an invitation in the phone app.
- **Donate.** Donors register once, set when they can be contacted, accept invitations and message the hospital, without giving away their phone number.

## Two apps

| APP           | WHO USES IT                                                                                   | WHERE                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Staff web app | Blood bank managers, hospital transfusion leads, donor coordinators, administrators, auditors | A computer browser, at <code>/staff</code> . Works on tablets and phones too.                                 |
| Phone app     | Donors and people requesting blood for a patient                                              | A phone browser, at the main address ( <code>/app</code> ). Can be installed on the home screen like any app. |

## Roles

What you see depends on your role. Staff are linked to one or more sites; LifeGrid checks your role and sites on every action, so menus only show what you can use.

### Blood bank manager

Sees the whole region. Approves transfers, dispatches them, starts donor appeals, manages blood bank stock.

### Hospital transfusion lead

Manages stock at their own hospital, confirms blood requests, receives transfers, records donor outcomes.

**Donor coordinator**

Looks after donors: deferrals, group verification, appeals, outcomes. Never sees donor contact details.

**Administrator**

Manages users, sites, settings and compatibility rules. Signs in with an authenticator code.

**Auditor**

Read-only access to everything, including the tamper-evident audit log.

**Donor · Requester**

Use the phone app. Sign in with a one-time code sent to their phone.

**Privacy by design**

LifeGrid stores only what its rules need. Donors have no name in the system, their area is rounded to about 1 km, and no medical detail is kept. Phone numbers are encrypted and shown to the other person in a conversation only when both choose to share them.

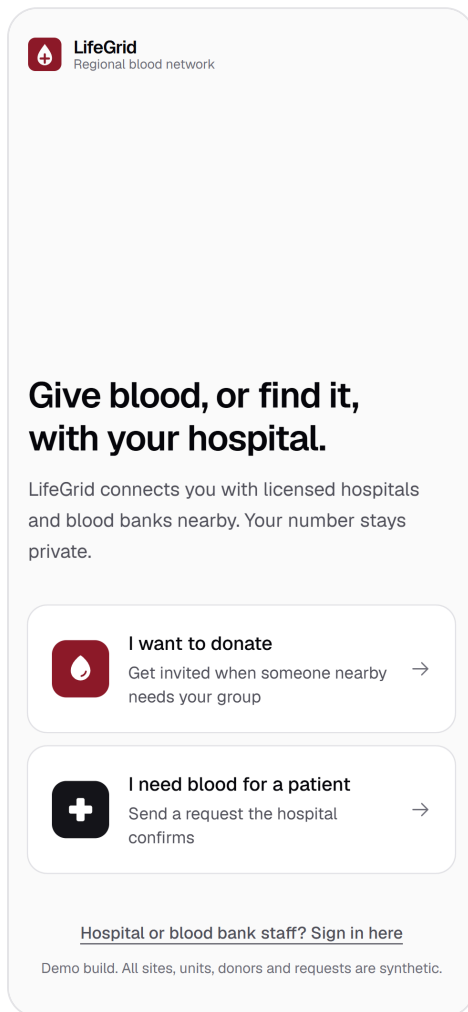
**Safety boundaries**

LifeGrid only recommends. A person approves every transfer, a hospital confirms every request, and donations happen only at licensed facilities. There are no payments to donors.

# Getting started

Open the LifeGrid address your organisation gives you. The main address opens the phone app; staff use the link at the bottom of it, or go straight to `/staff`.

## Signing in on the phone app



**Welcome screen.** Choose why you are here.

- 1 Tap **I want to donate** or **I need blood for a patient**.
- 2 Enter your mobile number with the country code, for example `+880 17...`, and tap **Send code**.
- 3 Type the 6-digit code from the text message and tap **Continue**. The first time, this creates your account; there is no password to remember.

### Codes expire

A code is valid for 5 minutes. After 5 wrong attempts the number is locked for 15 minutes, and you can ask for at most 3 codes an hour.

**Install the app.** If your phone supports it, an **Install the app** button appears under the two choices. It adds LifeGrid to your home screen so it opens like any other app. You can also use your browser's "Add to Home screen".

 **LifeGrid**  
Regional blood network

[← Back](#)

## Your phone number

We send a 6-digit code by SMS. No password needed.

Phone number

Include the country code.

Send code

[Hospital or blood bank staff? Sign in here](#)  
Demo build. All sites, units, donors and requests are synthetic.

Enter your number.

 **LifeGrid**  
Regional blood network

[← Back](#)

## Enter the code

Sent to +8801000009999. It expires in 5 minutes.

6-digit code

Development: the code appears in the API log (mock SMS).

Development code (no SMS sent): **006110**

Continue

[Hospital or blood bank staff? Sign in here](#)  
Demo build. All sites, units, donors and requests are synthetic.

**Enter the code.** In the demo build no text message is sent; the code appears on screen and you can tap it to fill it in.

## Signing in as staff

The image is a composite of two parts. The left part is a staff sign-in page for 'LifeGrid', a regional blood network. It features a logo at the top left, a 'STAFF SIGN-IN' header, and a 'Welcome back' message. Below this, there are input fields for 'Work email' (containing 'name@hospital.example...') and 'Password'. A red 'Sign in' button with a right arrow is positioned below the password field. At the bottom left, there is a link for 'Demo accounts'. The right part of the image is a dark-themed network diagram. It shows 14 sites connected by lines, with a central red node highlighted. Text at the bottom right explains that this is a live view of every unit in the region, where a blood bag about to expire is being transferred to a patient across town, and notes that 14 sites are connected and transfers always need a person's approval.

**Staff sign-in.** Work email and password. The link at the bottom leads to the phone app.

- 1 Enter your work email and password (at least 12 characters) and select **Sign in**.
- 2 If your account uses an authenticator app (always the case for administrators), enter the 6-digit code it shows and select **Verify and continue**.
- 3 LifeGrid opens the screen that fits your role: managers see the network overview, hospital leads their site's stock, coordinators the donor list, auditors the audit log.

The figure consists of four sequential screenshots of the LiferGid web application interface:

- Screenshot 1 (Leftmost):** Displays the 'Welcome back' message. Below it, a message states: 'Sign in with your work email. Administrators confirm with an authenticator code.' There are two input fields: 'Work email' (containing 'nablala.karm@lifergid.test') and 'Password' (with a masked input). A 'Sign in →' button is at the bottom.
- Screenshot 2:** Shows the 'Demo accounts' section. It includes a message: 'Pick a needed account to fill the form. The administrator needs a real authenticator code from the account printed to make sure.' Below this is a table of accounts:

| Account                        | Role                      |
|--------------------------------|---------------------------|
| nablala.karm@lifergid.test     | Blood bank manager        |
| hospita.1.karm@lifergid.test   | Hospital translation lead |
| hospita.2.karm@lifergid.test   | Hospital translation lead |
| hospita.3.karm@lifergid.test   | Hospital translation lead |
| hospita.4.karm@lifergid.test   | Hospital translation lead |
| hospita.5.karm@lifergid.test   | Hospital translation lead |
| hospita.6.karm@lifergid.test   | Hospital translation lead |
| hospita.7.karm@lifergid.test   | Hospital translation lead |
| hospita.8.karm@lifergid.test   | Hospital translation lead |
| hospita.9.karm@lifergid.test   | Hospital translation lead |
| hospita.10.karm@lifergid.test  | Hospital translation lead |
| hospita.11.karm@lifergid.test  | Hospital translation lead |
| hospita.12.karm@lifergid.test  | Hospital translation lead |
| hospita.13.karm@lifergid.test  | Hospital translation lead |
| hospita.14.karm@lifergid.test  | Hospital translation lead |
| hospita.15.karm@lifergid.test  | Hospital translation lead |
| hospita.16.karm@lifergid.test  | Hospital translation lead |
| hospita.17.karm@lifergid.test  | Hospital translation lead |
| hospita.18.karm@lifergid.test  | Hospital translation lead |
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| hospita.22.karm@lifergid.test  | Hospital translation lead |
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| hospita.73.karm@lifergid.test  | Hospital translation lead |
| hospita.74.karm@lifergid.test  | Hospital translation lead |
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| hospita.97.karm@lifergid.test  | Hospital translation lead |
| hospita.98.karm@lifergid.test  | Hospital translation lead |
| hospita.99.karm@lifergid.test  | Hospital translation lead |
| hospita.100.karm@lifergid.test | Hospital translation lead |

- Screenshot 3:** Shows the 'Authenticator code' screen. It says: 'Open your authenticator app and enter the 6-digit code for LiferGid.' There is a '6-digit code' input field (containing '0 0 0 0 0 0') and a 'Codes refresh every 30 seconds.' message. Below the input field is a 'Verify and continue' button.
- Screenshot 4 (Rightmost):** This is a duplicate of the second screenshot, showing the 'Demo accounts' list.

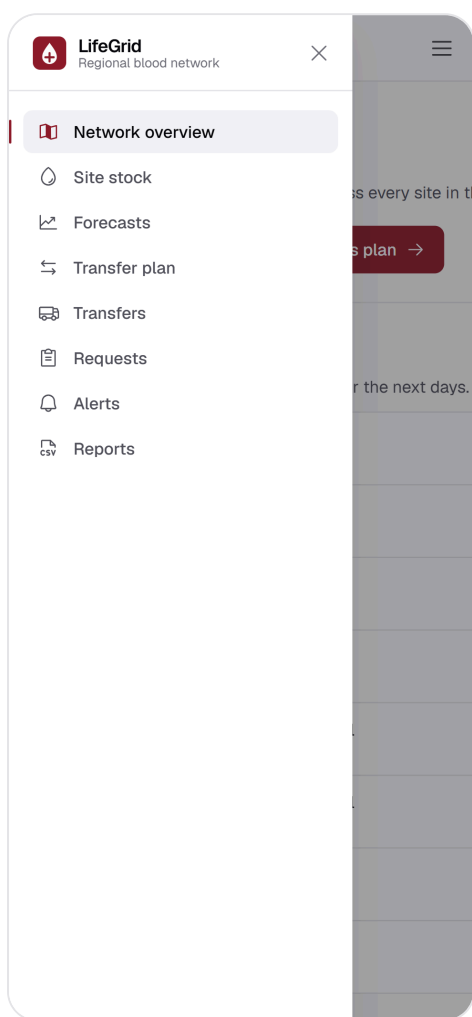
**Demo accounts.** In the demo build, open **Demo accounts** and pick a role to fill in the form.

**Authenticator step** for administrators.

### Staying signed in

You stay signed in for up to 7 days of use; LifeGrid renews your session in the background. Select **Sign out** at the bottom of the menu to end it. If an administrator disables your account, you are signed out at once. One browser holds one signed-in person; to try two roles at the same time, use a second browser or a private window.

## Staff app on a small screen



On a narrow screen the side menu folds away. Tap the menu button at the top right to open it; tap outside it or press **Esc** to close it. Tables that are wider than the screen scroll sideways.

Menu on a phone.

# For donors

Register once, keep your availability up to date, and LifeGrid will invite you only when a hospital near you needs your blood group and you are able to donate.

## Registering

The first time you sign in as a donor, LifeGrid asks for a short profile.

**LifeGrid** [Sign out]

STEP 2 OF 2

### Set up your donor profile

We only store what eligibility rules need. No name, no address, no medical history.

**Your blood group**

☐ O+
 ☐ O-
 ☐ A+
 ☒ A-
 ☐ B+
 ☐ B-
 ☐ AB+
 ☐ AB-

A coordinator can verify it later from a facility record.

**Your area**

Area set near 23.79, 90.40 (rounded to about 1 km)

**Birth year**

Used for the 18–60 age rule only.

☒ I weigh at least 50 kg

**Consent v1.2**

- LifeGrid stores your blood group, approximate area, birth year, and donation and deferral dates.
- Hospitals see only your group and area until you accept an invitation.
- Your phone number is shared only if you and the requester both choose to share it.
- Donations happen only at licensed facilities. No payment is involved.
- You can download or delete your data at any time.

☒ I agree to these terms

Home   Invitations   My data

- 1 Your blood group.** Tap your group, or **I don't know my group**. A coordinator can mark it as verified later from a facility record.
- 2 Your area.** Tap the map near where you live or work. LifeGrid rounds it to about 1 km before saving it; your exact address is never stored.
- 3 Birth year.** Used only for the age rule (18–60 by default).
- 4 Weight.** Tick the box if you weigh at least 50 kg.
- 5 Consent.** Read the consent text and tick **I agree to these terms**. You can't register without it.
- 6 Tap Save and continue.**

### What hospitals see

Until you accept an invitation, hospitals see only your blood group and area. Coordinators see a reference like `D-4F2A9C`, never your phone number.

Donor registration, filled in.

## Your home screen

The screenshot shows the LifeGrid app interface. At the top, there's a header with the LifeGrid logo and a 'Sign out' button. Below this is a 'DONATION STATUS' card with a dark background. It says 'You can donate' and 'You may be invited when someone nearby needs your group.' It also shows 'Group: O+ Self-reported' and 'Last donation: Never'. Below the card is a red bar indicating '1 open invitation'. Underneath is the 'Availability' section with a description 'Set to unavailable or travelling to pause invitations.' and three buttons: 'Available' (selected), 'Unavailable', and 'Travelling'. Below that is the 'Quiet hours' section with a description 'No invitations during these hours.' and two time pickers for 'From' and 'To'. There's a checkbox 'Still contact me during quiet hours for emergencies' which is checked. At the bottom of this section is a dropdown for 'Invitations per month, at most' set to '2'. The bottom navigation bar has three icons: 'Home' (selected), 'Invitations', and 'My data'.

Donor home.

**Donation status** tells you whether you can donate now. After a donation you have to wait 120 days (the default); after a deferral, until the date a coordinator set. While you wait, the card shows the date you become eligible again, and LifeGrid sends you one reminder on that day.

**Open invitations** appear as a red bar. Tap it to see them.

**Availability** pauses invitations without deleting anything:

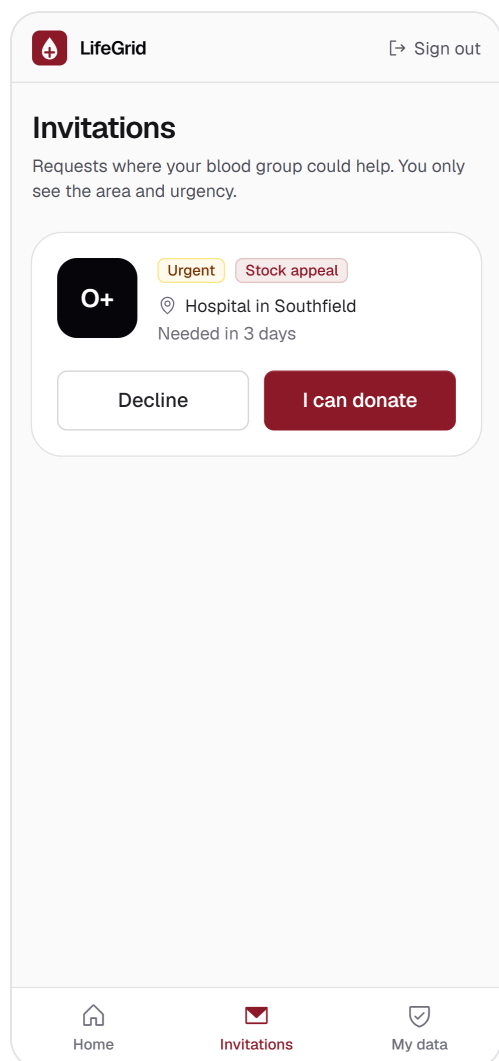
- **Available:** you can be invited.
- **Unavailable or Travelling:** no invitations.

**Quiet hours** stop invitations during the hours you choose, for example 22:00 to 07:00. Tick **Still contact me during quiet hours for emergencies** if you want emergency requests to reach you anyway.

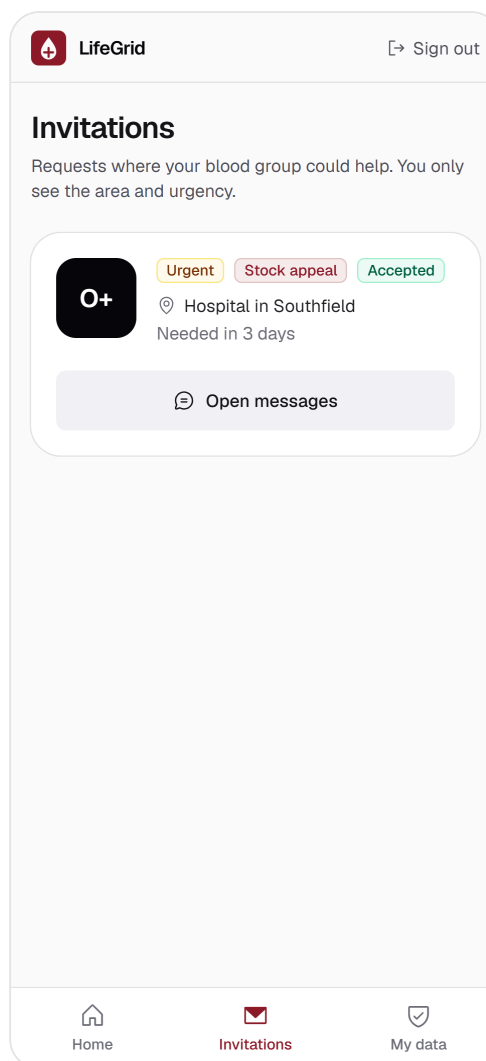
**Invitations per month** caps how often LifeGrid asks you (2 by default).

Changes save as soon as you make them.

## Answering an invitation



**An invitation.** Blood group needed, the area, urgency and when it is needed. A **Stock appeal** label means a hospital is running low rather than one patient.



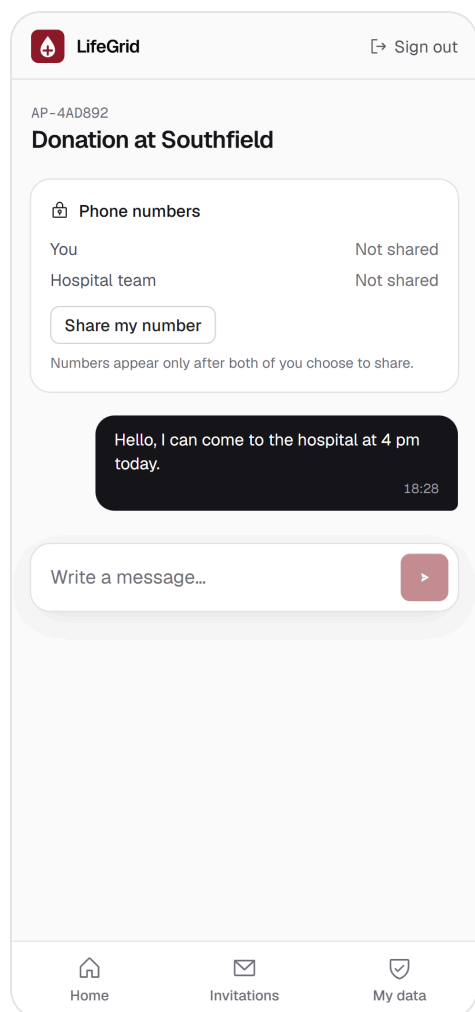
**After accepting.** The card shows **Accepted** and a button to open messages.

- 1 You get a text message: “LifeGrid: urgent need for O+ blood near Southfield...”. Open the app and tap **Invitations**.
- 2 Tap **I can donate** if you can go, or **Decline** if not. Declining is fine and does not count against you.
- 3 If you accept, the hospital is told and a private conversation opens.

### Why you were invited

LifeGrid invites donors whose group is compatible, who are available, past their waiting period, not deferred, within the search distance and outside their quiet hours. It invites a few people at a time, in waves, and stops as soon as enough have accepted, so nobody is asked more than needed.

## Messages and phone numbers



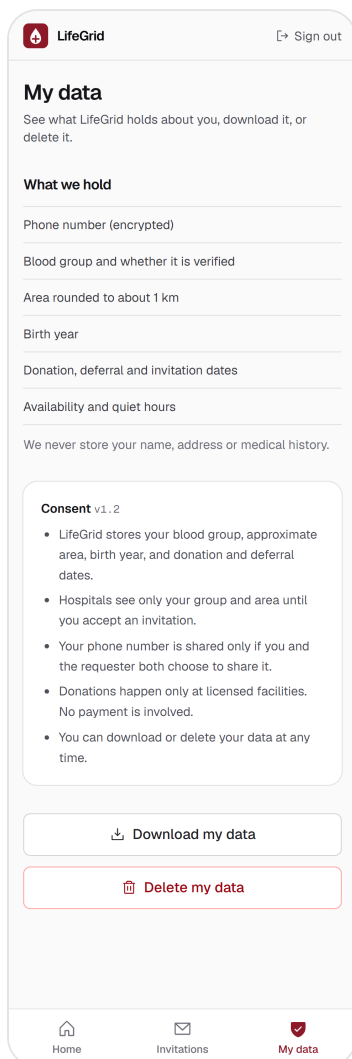
Conversation with the hospital team.

Use the conversation to agree on a time and place. Messages stay inside LifeGrid.

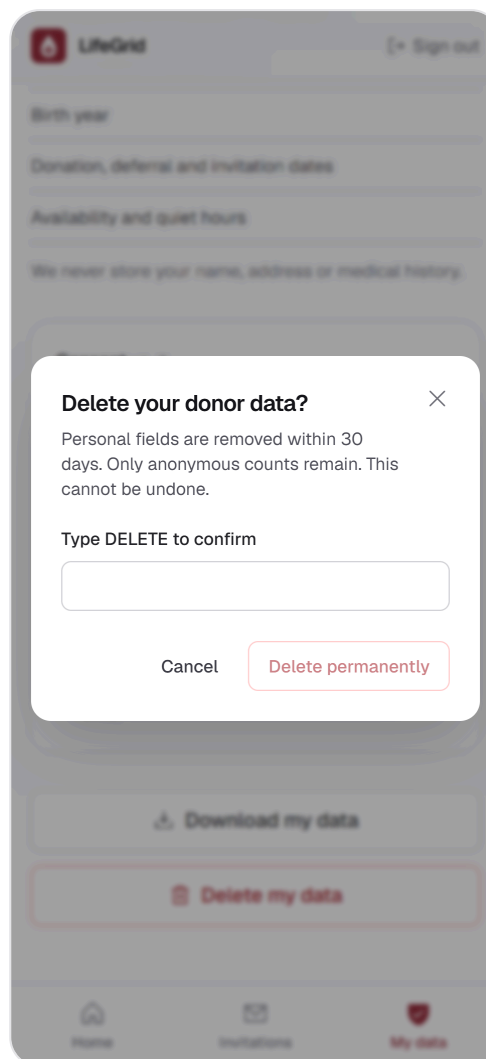
**Phone numbers** stay hidden. If you want to talk by phone, tap **Share my number**. Numbers appear for both of you only after *both* sides choose to share.

Donations happen only at the hospital or blood bank. Never hand over blood privately, and never accept payment.

## Your data



**My data.** What LifeGrid holds, the consent text, download and delete.



**Deleting.** Type DELETE to confirm.

- **Download my data** saves a file with your profile and invitation history.
- **Delete my data** removes your group, area, birth year, dates and phone number, and signs you out. Only anonymous counts remain, so the hospital's statistics stay correct. This can't be undone.

# Requesting blood for a patient

Use this when a patient you are helping needs blood. The hospital confirms every request before anything happens, then LifeGrid looks for stock first and asks donors only for what is missing.

## Making a request

**LifeGrid** → Sign out

### New request

The hospital confirms every request before anything happens.

**Patient blood group**

O+ O- A+ A- **B+** B- AB+ AB-

**Component**

Red cells

**Units needed**

4

1 to 20 units

**Hospital**

Southfield General Hospital

**Urgency**

Emergency **Urgent** Routine

**Needed by**

12/01/2026 06:00 PM

A transfusion lead at the hospital confirms every request before stock is reserved or donors are contacted.

**Submit request**

My requests **New request**

- 1 Sign in with **I need blood for a patient**. Your list of requests opens; tap **New request**.
- 2 **Patient blood group**: tap the group on the patient's record.
- 3 **Component**: red cells unless the doctor says platelets or plasma.
- 4 **Units needed**: 1 to 20.
- 5 **Hospital**: where the patient is treated.
- 6 **Urgency**: emergency, urgent or routine. Emergency invites donors faster and over a smaller distance first.
- 7 **Needed by**: the date and time the blood is needed.
- 8 Tap **Submit request**.

### Limits

You can have up to 3 open requests and create up to 5 a day. The hospital and needed-by time are required.

New request.

## Following your request

LifeGrid

→ Sign out

B+

RQ-466401

Awaiting hospital

The hospital needs to confirm this request before anything happens.

Awaiting hospital

Urgent

Units secured0 / 4

0

0

0

from stockdonors notifieddonors accepted

Submitted

Confirmed

Sourcing

Closed

HospitalSouthfield General Hospital

ComponentRed cells

Needed byDec 1, 18:00

Cancel request

My requests

New request

**Waiting for the hospital.** Nothing happens until a transfusion lead confirms.

LifeGrid

→ Sign out

B+

RQ-466401

Covered by stock

Network stock covers this request. No donors were contacted.

Covered by stock

Urgent

Units secured4 / 4

4

0

0

from stockdonors notifieddonors accepted

✓ Submitted

✓ Confirmed

Sourcing

Closed

HospitalSouthfield General Hospital

ComponentRed cells

Needed byDec 1, 18:00

Cancel request

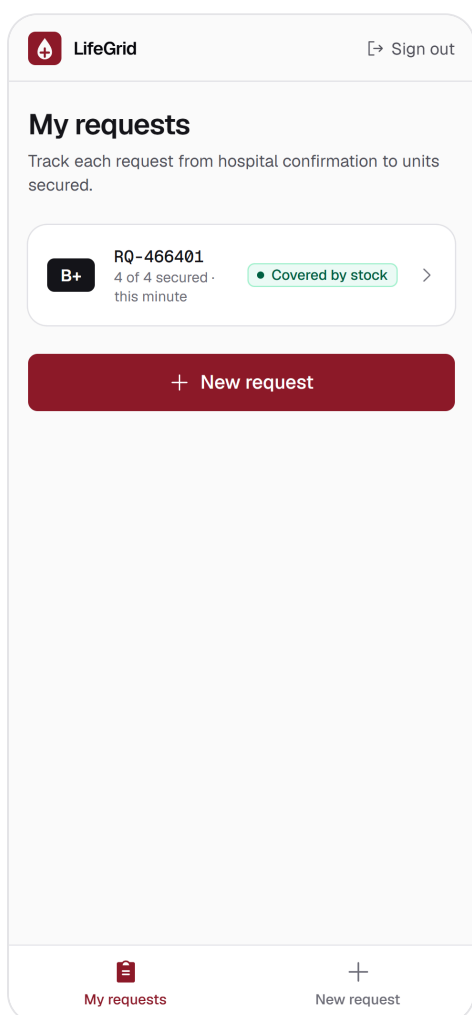
My requests

New request

**Confirmed and covered by stock.** Units secured fills up as stock is reserved or donors donate.

The status page shows units secured, how many came from stock, how many donors were notified and how many accepted, plus a four-step progress bar. It refreshes by itself.

| STATUS            | WHAT IT MEANS                                                                            |
|-------------------|------------------------------------------------------------------------------------------|
| Awaiting hospital | The hospital still has to confirm the request.                                           |
| Confirmed         | Confirmed; stock is being checked.                                                       |
| Covered by stock  | Units in the network cover the whole request. No donors were contacted.                  |
| Finding donors    | Stock didn't cover everything, so eligible donors nearby are being invited for the rest. |
| Fulfilled         | All units were secured.                                                                  |
| Partly fulfilled  | The needed-by time passed with some units secured.                                       |
| Unfilled          | The needed-by time passed with no units secured.                                         |
| Cancelled         | You or the hospital withdrew the request. Reserved units are released.                   |



**My requests**, with each request's status.

When donors accept, a **Donors who accepted** list appears on the status page. Tap a donor to message them; phone numbers are shared only if you both choose to share.

**My requests** lists everything you have asked for, newest first. Tap a request to open its status page.

To withdraw a request, open it, tap **Cancel request** and confirm. Any units reserved for it go back to stock and donor invitations stop.

# For blood bank managers

You see the whole region. Your daily work: check stock and low-stock warnings, review the overnight transfer plan, approve and dispatch transfers, and appeal to donors when a group runs low.

## Network overview

LifeGrid

Regional blood network

Network overview

Site stock

Forecasts

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Reports

nabila.karim@lifegrid.test

Blood bank manager

Sign out

Demo build. All sites, units, donors and requests are synthetic.

OCT 3, 2026

Network overview

Live stock, expiry pressure and alerts across every site in the region.

Run plan now

Review tonight's plan

Low stock

Red-cell groups below forecast demand for the next days. Appeal to eligible donors nearby.

A+

Harbourside Teaching Hospital

1 on hand - 3 needed in 2 days

Appeal to donors

A+

Elmhurst District Hospital

3 on hand - 5 needed in 2 days

Appeal to donors

B+

Elmhurst District Hospital

1 on hand - 3 needed in 2 days

Appeal to donors

B-

Greenway General Hospital

1 on hand - 2 needed in 2 days

Appeal to donors

A+

Meadowbrook General Hospital

8 on hand - 9 needed in 2 days

Appeal to donors

B-

Meadowbrook General Hospital

1 on hand - 2 needed in 2 days

Appeal to donors

AB-

Harbourside Teaching Hospital

0 on hand - 1 needed in 2 days

Appeal to donors

AB+

Bayview Teaching Hospital

3 on hand - 4 needed in 2 days

Appeal to donors

RED CELLS AVAILABLE

757

across 14 sites

EXPIRING IN 2 DAYS

0

move or use first

PROJECTED STOCKOUTS

0

within 2 days

OPEN ALERTS

0

0 critical

Sites by risk

Critical Watch Stable

No open alerts

New alerts appear here as soon as a rule fires.

Available stock by site and group

Bar shows how much of each cell expires soon. Red numbers mean none on hand.

Red cells

Platelets

Plasma

| SITE                                | O+  | O- | A+ | A- | B+ | B- | AB+ | AB- |
|-------------------------------------|-----|----|----|----|----|----|-----|-----|
| Riverside Regional Blood Bank BB-01 | 185 | 0  | 79 | 15 | 28 | 0  | 53  | 2   |
| Lakeview Regional Blood Bank BB-02  | 34  | 3  | 32 | 6  | 64 | 0  | 28  | 2   |
| Southfield General Hospital H-01    | 16  | 3  | 12 | 1  | 6  | 2  | 3   | 1   |
| Eastbank Medical Centre H-02        | 6   | 1  | 3  | 1  | 3  | 1  | 2   | 1   |
| Westwood District Hospital H-03     | 13  | 2  | 8  | 2  | 9  | 2  | 3   | 1   |
| Hillcrest Teaching Hospital H-04    | 5   | 1  | 5  | 1  | 3  | 1  | 1   | 1   |
| Meadowbrook General Hospital H-05   | 11  | 2  | 8  | 1  | 7  | 1  | 3   | 1   |
| Cedar Park Medical Centre H-06      | 6   | 0  | 5  | 0  | 3  | 1  | 2   | 0   |
| Old Town District Hospital H-07     | 11  | 2  | 11 | 2  | 9  | 1  | 3   | 0   |
| Harbourside Teaching Hospital H-08  | 3   | 1  | 1  | 1  | 2  | 1  | 0   | 0   |
| Greenway General Hospital H-09      | 10  | 2  | 7  | 2  | 8  | 1  | 1   | 1   |
| Stonebridge Medical Centre H-10     | 4   | 1  | 4  | 1  | 3  | 0  | 1   | 1   |
| Elmhurst District Hospital H-11     | 5   | 1  | 3  | 1  | 1  | 1  | 1   | 1   |
| Bayview Teaching Hospital H-12      | 15  | 2  | 13 | 1  | 8  | 0  | 3   | 0   |

0-2 days left

3-7 days left

8+ days left

Network overview, the manager's home screen.

- **Low stock** lists red-cell groups where a hospital holds fewer units than it is forecast to use in the next 2 days, worst first. Each row has **Appeal to donors** (see **Appealing to donors**).

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- **Key figures:** units available, units expiring within 2 days, projected stockouts and open alerts.
- **Sites by risk:** every hospital and blood bank on the map, coloured critical, watch or stable. Click a site for its counts and a link to its stock.
- **Available stock by site and group:** one row per site, one column per blood group. The small bar under each number shows how much expires within 2 days (red), in 3–7 days (amber) or later (grey). Red numbers mean none on hand. Switch between red cells, platelets and plasma at the top right.
- **Run plan now** recalculates the transfer plan immediately; **Review tonight's plan** opens it.

## Appealing to donors when stock runs low

### Appeal for A+ at Harbourside Teaching Hospital

Eligible donors nearby get an SMS and an in-app invitation in waves until enough accept. They see the group, the area and the urgency only.

Units needed

Projected shortfall: 2

Urgency

Emergency
Urgent
Routine

Cancel
Send appeal

### Appeal for A+ at Harbourside Teaching Hospital

Eligible donors nearby get an SMS and an in-app invitation in waves until enough accept. They see the group, the area and the urgency only.

✓ Appeal AP-48FEEC started. 6 donors invited in the first wave.

Done

**Appeal started**, with the first wave of invitations sent.

**Appeal dialog.** Units and urgency.

- 1 In **Low stock**, select **Appeal to donors** on the group and hospital you want to cover.
- 2 Check **Units needed** (the projected shortfall is filled in) and choose the **urgency**.
- 3 Select **Send appeal**. LifeGrid invites eligible donors near that hospital in waves, by text message and in the app, until enough accept. Donors see the group, the area and the urgency only.
- 4 The row now shows **Appeal running** ; you can't start a second appeal for the same group and hospital while one is open.

### How waves work

Each wave invites just enough donors to expect the shortfall plus 50% to accept. If not enough accept within the wait (10 minutes for emergency, 30 for urgent, 2 hours for routine), the next wave searches a wider area. If nobody accepts after the last wave, coordinators and bank managers are notified.

Follow an appeal on the **Requests** screen: appeals are marked **Donor appeal** . Open one to see which donors were invited and who accepted, and select **Message** next to a donor who accepted to talk to them.

AP-4AD892

Finding donors

×

3 units of Red cells · Southfield General Hospital · Needed Oct 6, 18:28

O+

Urgent

Units secured

0 / 3

|                 |                      |                      |
|-----------------|----------------------|----------------------|
| 0<br>from stock | 7<br>donors notified | 1<br>donors accepted |
|-----------------|----------------------|----------------------|

✓ Submitted

✓ Confirmed

Sourcing

Closed

7 donors invited

|    |                 |         |          |
|----|-----------------|---------|----------|
| O+ | D-65A597 wave 1 | Message | Accepted |
| O+ | D-D604DE wave 1 |         | Invited  |
| O- | D-EDCACE wave 1 |         | Invited  |
| O+ | D-34EB15 wave 1 |         | Invited  |
| O+ | D-829A37 wave 1 |         | Invited  |
| O+ | D-569FFA wave 1 |         | Invited  |

**Appeal details:** progress and the donors invited, by reference and group only.

Messages with donor

×

AP-4AD892

Hello, I can come to the hospital at 4 pm today.  
18:28

Thank you! The donation desk is on the ground floor.  
18:28

Write a message...

Send message

**Messaging a donor who accepted.**

## Site stock

 LifeGrid  
Regional blood network

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Blood bank manager

Sign out

Demo build. All sites, units, donors and requests are synthetic.

BB-01 - BLOOD BANK

### Riverside Regional Blood Bank

Units at this site, first-expired first. Select units to issue, quarantine or discard.

Riverside Regional Blood Bank

Import CSV

Receive unit

Status: Available

Component: All

Group: All

Expiry: All

375 units · sorted first-expired, first-out

| <input type="checkbox"/> | UNIT         | GROUP | COMPONENT | EXPIRES      | REMAINING | STATUS    |
|--------------------------|--------------|-------|-----------|--------------|-----------|-----------|
| <input type="checkbox"/> | LG26-0000542 | AB+   | Red cells | Oct 15, 2026 | 11 days   | Available |
| <input type="checkbox"/> | LG26-0000544 | AB+   | Red cells | Oct 15, 2026 | 11 days   | Available |
| <input type="checkbox"/> | LG26-0000543 | AB+   | Red cells | Oct 15, 2026 | 11 days   | Available |
| <input type="checkbox"/> | LG26-0000507 | A-    | Red cells | Oct 20, 2026 | 16 days   | Available |
| <input type="checkbox"/> | LG26-0000548 | AB+   | Red cells | Oct 20, 2026 | 16 days   | Available |
| <input type="checkbox"/> | LG26-0000547 | AB+   | Red cells | Oct 20, 2026 | 16 days   | Available |
| <input type="checkbox"/> | LG26-0000508 | A-    | Red cells | Oct 20, 2026 | 16 days   | Available |
| <input type="checkbox"/> | LG26-0000551 | AB+   | Red cells | Oct 20, 2026 | 16 days   | Available |
| <input type="checkbox"/> | LG26-0000546 | AB+   | Red cells | Oct 20, 2026 | 16 days   | Available |
| <input type="checkbox"/> | LG26-0000545 | AB+   | Red cells | Oct 20, 2026 | 16 days   | Available |
| <input type="checkbox"/> | LG26-0000550 | AB+   | Red cells | Oct 20, 2026 | 16 days   | Available |
| <input type="checkbox"/> | LG26-0000549 | AB+   | Red cells | Oct 20, 2026 | 16 days   | Available |

**Site stock** for a blood bank. Units are sorted first-expired, first-out.

- **Choose the site** with the drop-down at the top right (managers can open any site).
- **Filter** by status, component, group and expiry band. Filters are kept in the address, so you can bookmark or share a filtered view.
- **Remaining** shows days left, or hours on the last day, in red when 2 days or fewer remain.
- Select a unit code to open its **history**.

## Changing units

LifeGrid  
Regional blood network

BB-01 · BLOOD BANK

### Riverside Regional Blood Bank

Units at this site, first-expired first. Select units to issue, quarantine or discard.

Riverside Regional Blood Bank Import CSV + Receive unit

Status: Available Component: All Group: All Expiry: All 375 units · sorted first-expired, first-out

| UNIT                                             | GROUP | COMPONENT | EXPIRES      | REMAINING | STATUS    |
|--------------------------------------------------|-------|-----------|--------------|-----------|-----------|
| <input checked="" type="checkbox"/> LG26-0000542 | AB+   | Red cells | Oct 15, 2026 | 11 days   | Available |
| <input checked="" type="checkbox"/> LG26-0000544 | AB+   | Red cells | Oct 15, 2026 | 11 days   | Available |
| <input type="checkbox"/> LG26-0000543            | AB+   | Red cells | Oct 15, 2026 | 11 days   | Available |
| <input type="checkbox"/> LG26-0000507            | A-    | Red cells | Oct 20, 2026 | 16 days   | Available |
| <input type="checkbox"/> LG26-0000548            | AB+   | Red cells | Oct 20, 2026 | 16 days   | Available |
| <input type="checkbox"/> LG26-0000547            | AB+   | Red cells | Oct 20, 2026 | 16 days   | Available |
| <input type="checkbox"/> LG26-0000508            | A-    | Red cells | Oct 20, 2026 | 16 days   | Available |
| <input type="checkbox"/> LG26-0000551            | AB+   | Red cells | Oct 20, 2026 | 16 days   | Available |
| <input type="checkbox"/> LG26-0000546            | AB+   | Red cells | Oct 20, 2026 | 16 days   | Available |
| <input type="checkbox"/> LG26-0000545            | AB+   | Red cells | Oct 20, 2026 | 16 days   | Available |
| <input type="checkbox"/> LG26-0000550            | AB+   | Red cells | Oct 20, 2026 | 16 days   | Available |
| <input type="checkbox"/> LG26-0000549            | AB+   | Red cells | Oct 20, 2026 | 16 days   | Available |

2 selected Quarantine Cancel

**Two units selected.** The bar at the bottom offers only actions that apply to every selected unit.

- 1 Tick the units. The action bar appears at the bottom.
- 2 Choose an action. Every action shows the unit count and the codes before anything changes.
- 3 For **Quarantine** and **Discard** you must give a reason.

### Quarantine 2 units?

This changes stock for 2 units and is written to the audit log.

LG26-0000542 AB+ LG26-0000544 AB+

Reason (required)

Fridge 2 temperature alarm at 06:10

Cancel Quarantine · 2

**Quarantine** needs a reason and confirms the count.

| ACTION           | FROM                                  | WHO                         |
|------------------|---------------------------------------|-----------------------------|
| Issue            | Reserved (for a confirmed request)    | Hospital lead               |
| Quarantine       | Available, reserved, in transit       | Hospital lead, bank manager |
| Clear quarantine | Quarantined                           | Bank manager                |
| Discard          | Quarantined (bank manager) or expired | Hospital lead, bank manager |

Units expire automatically at their expiry time; reservations not issued within 24 hours return to stock.

## Receiving and importing units

### Receive a unit

Record a released, labelled unit arriving at this site.

Unit code

ISBT 128 in production; the demo uses LG26-...

Group

O+ ▼

Component

Red cells ▼

Collected

10/02/2026 06:27 PM

Expires

11/13/2026 06:27 PM

Cancel

Receive unit

**Receive unit.** The expiry date is filled in from the component's shelf life; you can change it.

### Import units from CSV

Good rows are imported. Bad rows are reported by row number with the reason.

1 unit imported

2 rows rejected

| ROW | REASON                           |
|-----|----------------------------------|
| 3   | ABO group must be O, A, B or AB. |
| 4   | Expiry must be after collection. |

Done

**Import result.** Good rows are imported; bad rows are listed with the reason.

**Receive unit** records one released, labelled unit arriving at the site: unit code, group, component, collection and expiry. **Import CSV** takes a file (or pasted text) with the columns `unit_code`, `abo`, `rh`, `component`, `collected_at`, `expires_at`; select **Download template** for an example.

Unit history

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nabila.karim@lifegrid.test

Blood bank manager

Sign out

Demo build. All sites, units, donors and requests are synthetic.

← Riverside Regional Blood Bank

BLOOD UNIT

LG26-0000542

Quarantined

GROUP

AB+

COLLECTED

Sep 2, 09:00

REMAINING

11 days

COMPONENT

Red cells

EXPIRES

Oct 15, 00:00

SITE

Riverside Regional Blood Bank

Movement history

Available → Quarantined

Fridge 2 temperature alarm at 06:10  
Oct 3, 18:27 · nabila.karim@lifegrid.test

Available

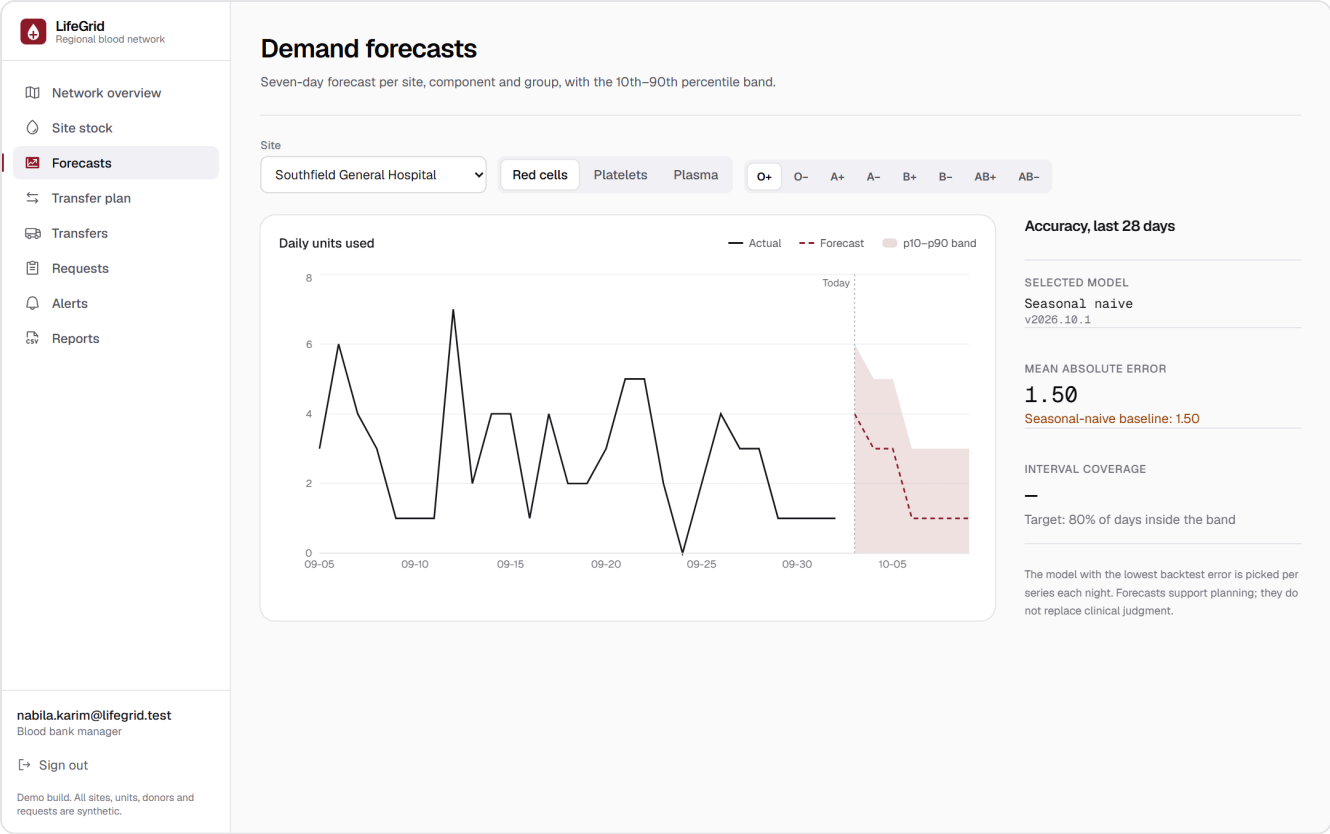
Received at site (demo seed)  
Sep 3, 10:00 · system

Unit detail: every status change with who did it, when and why.

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# Forecasts



**Demand forecast** for one hospital, component and group.

Choose a site, a component and a group. The chart shows the last 28 days of actual use (black), the forecast for the next 7 days (dashed red) and the likely range (shaded, 10th to 90th percentile). The panel on the right names the model chosen for this series, its error compared with the simple “same day last week” baseline, and how often actual use fell inside the range. Forecasts are recalculated every night at 01:30.

# Transfer plan

LifeGrid

Regional blood network

- Network overview
- Site stock
- Forecasts
- Transfer plan**
- Transfers
- Requests
- Alerts
- Reports

nabila.karim@lifegrid.test

Blood bank manager

Sign out

Demo build. All sites, units, donors and requests are synthetic.

GENERATED OCT 3, 18:27

Transfer plan

Recommended moves for the next 3 days. Nothing moves until a manager approves it.

Solver OPTIMAL 1.0 s

Projected effect

Units projected to expire unused, and demand projected to go unmet, over the plan horizon.

EXPIRED UNITS

Without 1

With plan 0

SHORT UNITS

Without 166

With plan 69

Recommendations

53 awaiting decision

|                                                                                                                 |                                                                                                                                 |                                          |
|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| <div>Lakeview Regional Blood Bank</div> <div>→ Southfield General Hospital</div> <div>Red cells · 18.0 km</div> | <div>A+</div> <div>9</div> <div>of 9</div> <div>Relay or swap between sites</div> <div>Avoids 0 expired and 0 short</div>       | <div>× Reject</div> <div>✓ Approve</div> |
| <div>Lakeview Regional Blood Bank</div> <div>→ Westwood District Hospital</div> <div>Red cells · 25.5 km</div>  | <div>O+</div> <div>8</div> <div>of 8</div> <div>Projected shortage at destination</div> <div>Avoids 0 expired and 5 short</div> | <div>× Reject</div> <div>✓ Approve</div> |
| <div>Riverside Regional Blood Bank</div> <div>→ Bayview Teaching Hospital</div> <div>Red cells · 27.2 km</div>  | <div>B+</div> <div>7</div> <div>of 7</div> <div>Projected shortage at destination</div> <div>Avoids 0 expired and 3 short</div> | <div>× Reject</div> <div>✓ Approve</div> |
| <div>Southfield General Hospital</div> <div>→ Meadowbrook General Hospital</div> <div>Red cells · 20.4 km</div> | <div>A+</div> <div>7</div> <div>of 7</div> <div>Projected shortage at destination</div> <div>Avoids 0 expired and 7 short</div> | <div>× Reject</div> <div>✓ Approve</div> |
| <div>Lakeview Regional Blood Bank</div> <div>→ Greenway General Hospital</div> <div>Red cells · 4.8 km</div>    | <div>O+</div> <div>6</div> <div>of 6</div> <div>Projected shortage at destination</div> <div>Avoids 0 expired and 6 short</div> | <div>× Reject</div> <div>✓ Approve</div> |
| <div>Bayview Teaching Hospital</div> <div>→ Elmhurst District Hospital</div> <div>Red cells · 6.5 km</div>      | <div>A+</div> <div>6</div> <div>of 6</div> <div>Projected shortage at destination</div> <div>Avoids 0 expired and 6 short</div> | <div>× Reject</div> <div>✓ Approve</div> |

**Transfer plan:** projected effect at the top, recommendations below.

Every night at 02:00 LifeGrid works out which transfers would cut both waste and shortages over the next 3 days, at low transport cost. **Projected effect** compares expired and short units with and without the plan.

Each recommendation shows the route, group, component, distance, reason and expected benefit:

- Expiry risk at source** : the units would expire unused where they are.
- Projected shortage at destination** : the destination is forecast to run short.
- Relay or swap between sites** : part of a chain that helps a third site.
- For a blood request** : proposed when a hospital confirmed a request and stock was elsewhere.

- Lower the unit count if you want to move fewer units.
- Select **Approve** (confirm in the dialog) or **Reject**. Nothing moves without approval.
- Approved lines appear under **Transfers**, ready to dispatch.

×

## Approve transfer of 2 units?

Lakeview Regional Blood Bank → Southfield General Hospital · A+ Red cells

You reduced this transfer from 9 to 2 units.

Cancel

Approve

**Approving** a recommendation with an edited count.

## Transfers

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Blood bank manager

Sign out

Demo build. All sites, units, donors and requests are synthetic.

Transfers

Dispatch approved transfers and receive incoming units. Location changes only on receipt.

To dispatch 1

In transit 0

Received 0

Lakeview Regional Blood Bank → Southfield General Hospital

A+

2 units

Approved

Dispatch

**Transfers to dispatch.** Tabs for to dispatch, in transit and received.

- 1 Select **Dispatch**. LifeGrid picks the actual units first-expired-first-out, only from units with enough shelf life left to travel (5 days for red cells by default).
- 2 Check the list and select **Dispatch**. The units become 

In transit

 ; their location changes only when the destination receives them.

Dispatch 2 units

Units picked first-expired, first-out among those with enough shelf life left to travel.

|   |              |         |
|---|--------------|---------|
| 1 | LG26-0000634 | 32 days |
| 2 | LG26-0000635 | 32 days |

Cancel

Dispatch 2 units

Dispatch with the picked units.

LifeGrid

Regional blood network

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Transfers

Dispatch approved transfers and receive incoming units. Location changes only on receipt.

To dispatch

In transit

Received

Lakeview Regional Blood Bank → Southfield General Hospital

A+ 2 units 15 minutes

Awaiting receipt at destination

nabila.karim@lifegrid.test

Blood bank manager

Sign out

Demo build. All sites, units, donors and requests are synthetic.

In transit, waiting for the destination to receive.

Requests

LifeGrid

Regional blood network

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Blood requests

Confirm requests for your site and follow stock coverage and donor matching.

All requests

Open

Closed

|    |           |        |                               |                |                  |
|----|-----------|--------|-------------------------------|----------------|------------------|
| B+ | RQ-466401 | Urgent | Southfield General Hospital   | 4 of 4 secured | Covered by stock |
| O+ | AP-4AD892 | Urgent | Southfield General Hospital   | 0 of 3 secured | Finding donors   |
| A+ | AP-48FEEC | Urgent | Harbourside Teaching Hospital | 0 of 2 secured | Finding donors   |

nabila.karim@lifegrid.test

Blood bank manager

Sign out

Demo build. All sites, units, donors and requests are synthetic.

Requests across the region, with donor appeals marked.

Managers see every request and appeal. Select one to see its progress, its donors and, for requests covered from other sites, a link to review the proposed transfers.

# For hospital transfusion leads

You look after stock at your hospital, confirm requests for your patients, receive transfers and record whether donors donated.

## Your hospital's stock

LifeGrid  
Regional blood network

Site stock

Forecasts

Transfers

Requests

Alerts

tanvir.rahman@lifegrid.test  
Hospital transfusion lead

Sign out

Demo build. All sites, units, donors and requests are synthetic.

H-01 - HOSPITAL

Southfield General Hospital

Units at this site, first-expired first. Select units to issue, quarantine or discard.

Import CSV

Receive unit

Status

Component

Group

Expiry

Available

All

All

All

86 units · sorted first-expired, first-out

| UNIT                                  | GROUP | COMPONENT | EXPIRES      | REMAINING | STATUS    |
|---------------------------------------|-------|-----------|--------------|-----------|-----------|
| <input type="checkbox"/> LG26-0000759 | A-    | Platelets | Oct 4, 2026  | 5 hours   | Available |
| <input type="checkbox"/> LG26-0000758 | O+    | Platelets | Oct 4, 2026  | 5 hours   | Available |
| <input type="checkbox"/> LG26-0000760 | B-    | Platelets | Oct 4, 2026  | 5 hours   | Available |
| <input type="checkbox"/> LG26-0000761 | AB-   | Platelets | Oct 5, 2026  | 1 day     | Available |
| <input type="checkbox"/> LG26-0000033 | B+    | Red cells | Oct 22, 2026 | 18 days   | Available |
| <input type="checkbox"/> LG26-0000034 | B+    | Red cells | Oct 22, 2026 | 18 days   | Available |
| <input type="checkbox"/> LG26-0000039 | B-    | Red cells | Oct 25, 2026 | 21 days   | Available |
| <input type="checkbox"/> LG26-0000036 | B+    | Red cells | Oct 26, 2026 | 22 days   | Available |
| <input type="checkbox"/> LG26-0000001 | O+    | Red cells | Oct 26, 2026 | 22 days   | Available |
| <input type="checkbox"/> LG26-0000035 | B+    | Red cells | Oct 26, 2026 | 22 days   | Available |
| <input type="checkbox"/> LG26-0000044 | AB-   | Red cells | Oct 28, 2026 | 24 days   | Available |
| <input type="checkbox"/> LG26-0000020 | A-    | Red cells | Oct 28, 2026 | 24 days   | Available |

**Site stock** opens on your hospital. Leads see only their own sites' units; other sites appear as totals on the overview.

Stock actions work as described in **Site stock**. You can issue reserved units to patients, quarantine and discard.

## Confirming a request

**Requests:** anything waiting for you is highlighted at the top.

**Before confirming.**

**After confirming:** here stock covered it all.

- 1 Open the request under **needs confirmation** and check it against the patient's record.
- 2 Select **Confirm request**. LifeGrid then:
  - reserves compatible units at your hospital, best match first, then first-expired-first-out;
  - proposes transfers from other sites for anything still missing (a bank manager approves them);

- invites donors only for the units stock can't cover (red cells). Platelet and plasma shortfalls go to the coordinators instead.

- 3 The requester sees each step on their phone. Use **Cancel request** if it is no longer needed; reserved units go back to stock.

#### Your own requests

Requests you create with **New request** on this screen are confirmed for your hospital straight away.

## Recording donor outcomes

When a donor who accepted comes in (or doesn't), open the request, find the donor in the list and select **Donated**, **No-show** or **Deferred on site**. A donation counts towards units secured and resets that donor's waiting period. Outcomes are the only thing that changes a donor's reliability score. The screenshot in [chapter 7](#) shows the buttons.

## Receiving transfers

On **Transfers**, the **In transit** tab lists transfers coming to your hospital. When the box arrives, select **Receive** and confirm the unit codes. The units become available at your hospital.

# For donor coordinators

You keep the donor list accurate and safe: deferrals, verified groups, appeals and outcomes. You never see donors' names or contact details.

## Donor list

LifeGrid  
Regional blood network

Forecasts

Requests

**Donors**

Alerts

imran.chowdhury@lifegrid.test  
Donor coordinator

Sign out

Demo build. All sites, units, donors and requests are synthetic.

Donors

Eligibility, deferrals and group verification for donors in your region.

Contact details are never shown here

Low stock

Red-cell groups below forecast demand for the next days. Appeal to eligible donors nearby.

A+

Harbourside Teaching Hospital

1 on hand - 3 needed in 2 days

Appeal running

A+

Elmhurst District Hospital

3 on hand - 5 needed in 2 days

Appeal to donors

B+

Elmhurst District Hospital

1 on hand - 3 needed in 2 days

Appeal to donors

B-

Greenway General Hospital

1 on hand - 2 needed in 2 days

Appeal to donors

A+

Meadowbrook General Hospital

8 on hand - 9 needed in 2 days

Appeal to donors

B-

Meadowbrook General Hospital

1 on hand - 2 needed in 2 days

Appeal to donors

AB-

Harbourside Teaching Hospital

0 on hand - 1 needed in 2 days

Appeal to donors

AB+

Bayview Teaching Hospital

3 on hand - 4 needed in 2 days

Appeal to donors

Search

Donor ref or area...

Group

All

▼

Eligibility

All

▼

401 donors

| DONOR    | GROUP | AREA (=1 KM) | LAST DONATION | ELIGIBILITY  | AVAILABILITY | RELIABILITY |              |       |
|----------|-------|--------------|---------------|--------------|--------------|-------------|--------------|-------|
| D-D13F1B | A-    | Southfield   | Never         | Eligible now | Available    | 0.50        | Verify group | Defer |

Donors, with the low-stock panel at the top.

- Search by donor reference, filter by group and eligibility.
- A solid group chip with a check mark means the group is verified; a dashed chip means self-reported.
- **Eligibility** shows Eligible now, a deferral date, or the date the waiting period ends.
- **Reliability** is built from past responses and donations (0 to 1).

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## Deferrals and group verification

Record deferral for D-D13F1B

×

Only the category and date are stored. No medical detail.

Category

Low haemoglobin

Eligible again on

12/15/2026

Cancel

Save deferral

Record deferral.

Select **Defer**, pick a category and the date the donor can donate again. Only the category and date are stored, never medical detail. The donor is not invited until that date.

Select **Verify group** when you have confirmed a donor's group from a facility record. Verified donors are preferred when scores are tied.

## Appeals

Coordinators can start donor appeals from the same **Low stock** panel as bank managers; see [Appealing to donors](#).

Recording outcomes

AP-4AD892

Finding donors

×

3 units of Red cells · Southfield General Hospital · Needed Oct 6, 18:28

O+

Urgent

Units secured

0 / 3

0

from stock

7

donors notified

1

donors accepted

✓ Submitted

✓ Confirmed

Sourcing

Closed

7 donors invited

O+

D-65A597 wave 1

Accepted

Donated

No-show

Deferred on site

O+

D-D604DE wave 1

Invited

O-

D-EDCACE wave 1

Invited

O+

D-34EB15 wave 1

Invited

O+

D-829A37 wave 1

Invited

Request details for a coordinator: donors who accepted have Donated, No-show and Deferred on site buttons.

# For administrators and auditors

Administrators manage who can do what, and the rules LifeGrid follows. Every change is versioned and written to a tamper-evident audit log that auditors can check at any time.

## Users


**LifeGrid**  
Regional blood network

- Network overview
- Forecasts
- Alerts
- Reports
- Administration**
- Audit log

**sadia.haque@lifegrid.test**  
Administrator

[+] Sign out

Demo build. All sites, units, donors and requests are synthetic.

### Administration

Users, sites, settings and compatibility rules. Every change is versioned and audited.

Users
Sites
Settings
Compatibility

| WORK EMAIL                     | ROLE                      | SITE SCOPE   | AUTHENTICATOR | ACTIVE                              |
|--------------------------------|---------------------------|--------------|---------------|-------------------------------------|
| farhana.siddiqui@lifegrid.test | Hospital transfusion lead | H-03, H-02   | Off           | <input checked="" type="checkbox"/> |
| imran.chowdhury@lifegrid.test  | Donor coordinator         | Region-wide  | Off           | <input checked="" type="checkbox"/> |
| mahmud.alam@lifegrid.test      | Auditor                   | Region-wide  | Off           | <input checked="" type="checkbox"/> |
| nabila.karim@lifegrid.test     | Blood bank manager        | BB-01, BB-02 | Off           | <input checked="" type="checkbox"/> |
| sadia.haque@lifegrid.test      | Administrator             | Region-wide  | On            | <input checked="" type="checkbox"/> |
| tanvir.rahman@lifegrid.test    | Hospital transfusion lead | H-01         | Off           | <input checked="" type="checkbox"/> |

Disabling a user ends their sessions within 1 minute.

**Users:** role, site scope, authenticator and active switch.

- Change a user's **role** with the drop-down. Role and scope changes are audited.
- Turn **Active** off to disable an account; that person is signed out everywhere at once.
- Administrators must use an authenticator; the column shows Required until one is set up.

# Sites

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sadia.haque@lifegrid.test

Administrator

Sign out

Demo build. All sites, units, donors and requests are synthetic.

Administration

Users, sites, settings and compatibility rules. Every change is versioned and audited.

Users

Sites

Settings

Compatibility

| CODE  | NAME                          | TYPE       | LOCATION         | TIME ZONE  | ACTIVE |
|-------|-------------------------------|------------|------------------|------------|--------|
| BB-01 | Riverside Regional Blood Bank | Blood bank | 23.6918, 90.4389 | Asia/Dhaka | Active |
| BB-02 | Lakeview Regional Blood Bank  | Blood bank | 23.7475, 90.2965 | Asia/Dhaka | Active |
| H-01  | Southfield General Hospital   | Hospital   | 23.8893, 90.3816 | Asia/Dhaka | Active |
| H-02  | Eastbank Medical Centre       | Hospital   | 24.0015, 90.6517 | Asia/Dhaka | Active |
| H-03  | Westwood District Hospital    | Hospital   | 23.9766, 90.2935 | Asia/Dhaka | Active |
| H-04  | Hillcrest Teaching Hospital   | Hospital   | 23.4736, 90.5243 | Asia/Dhaka | Active |
| H-05  | Meadowbrook General Hospital  | Hospital   | 24.0271, 90.5145 | Asia/Dhaka | Active |
| H-06  | Cedar Park Medical Centre     | Hospital   | 23.7946, 90.0874 | Asia/Dhaka | Active |
| H-07  | Old Town District Hospital    | Hospital   | 24.0580, 90.1586 | Asia/Dhaka | Active |
| H-08  | Harbourside Teaching Hospital | Hospital   | 23.6245, 90.4361 | Asia/Dhaka | Active |
| H-09  | Greenway General Hospital     | Hospital   | 23.7222, 90.2588 | Asia/Dhaka | Active |
| H-10  | Stonebridge Medical Centre    | Hospital   | 23.6174, 90.7442 | Asia/Dhaka | Active |
| H-11  | Elmhurst District Hospital    | Hospital   | 23.8717, 90.6506 | Asia/Dhaka | Active |
| H-12  | Bayview Teaching Hospital     | Hospital   | 23.8159, 90.6688 | Asia/Dhaka | Active |

**Sites:** every hospital and blood bank with code, type, location, time zone and status.

# Settings


**LifeGrid**  
Regional blood network

- Network overview
- Forecasts
- Alerts
- Reports
- Administration**
- Audit log

**sadia.haque@lifegrid.test**  
Administrator  
[Sign out](#)  
Demo build. All sites, units, donors and requests are synthetic.

|                            |                                                                                            |    |              |
|----------------------------|--------------------------------------------------------------------------------------------|----|--------------|
| p_lock_minutes             | <input type="text" value="15"/>                                                            | v1 | Oct 3, 18:27 |
| p_max_attempts             | <input type="text" value="5"/>                                                             | v1 | Oct 3, 18:27 |
| p_requests_per_hour        | <input type="text" value="3"/>                                                             | v1 | Oct 3, 18:27 |
| p_ttl_minutes              | <input type="text" value="5"/>                                                             | v1 | Oct 3, 18:27 |
| ssword_min_length          | <input type="text" value="12"/>                                                            | v1 | Oct 3, 18:27 |
| an_horizon_days            | <input type="text" value="3"/>                                                             | v1 | Oct 3, 18:27 |
| an_run_time                | <input type="text" value="02:00"/>                                                         | v1 | Oct 3, 18:27 |
| atelets_allow_out_of_group | <input type="text" value="On"/>                                                            | v1 | Oct 3, 18:27 |
| atelets_prefer_rh_match    | <input type="text" value="On"/>                                                            | v1 | Oct 3, 18:27 |
| quester_daily_limit        | <input type="text" value="5"/>                                                             | v1 | Oct 3, 18:27 |
| quester_open_limit         | <input type="text" value="3"/>                                                             | v1 | Oct 3, 18:27 |
| seivation_timeout_hours    | <input type="text" value="24"/>                                                            | v1 | Oct 3, 18:27 |
| elf_life_days              | <input data-bbox="592 954 820 976" type="text" value='{"red_cells":42,"platelets":5,"p.'/> | v1 | Oct 3, 18:27 |
| lver_time_limit_s          | <input type="text" value="45"/>                                                            | v2 | Oct 3, 18:29 |

Settings with the change history on the right.

Business rules live here rather than in the software: shelf lives, minimum shelf life for transfers, expiry alert windows, waiting period after donation, age limits, invitation limits, wave timing and search distances, plan time and solver limits, request limits. Edit a value and select **Save**. Values with several parts (for example per component) are edited as text in the form `{"red_cells": 5, "platelets": 1, "plasma": 30}`. The history records the old and new value, who changed it and when.

## Clinical values

Shelf life, compatibility, eligibility and waiting periods are prototype defaults. A qualified clinician must review them against local standards before real use.

# Compatibility rules

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Sign out

Demo build. All sites, units, donors and requests are synthetic.

Administration

Users, sites, settings and compatibility rules. Every change is versioned and audited.

Users

Sites

Settings

Compatibility

Red cells

Platelets

Plasma

Placeholder clinical defaults. Clinician review required before real use.

Rows are recipients, columns are donor groups. Numbers are preference rank; 1 is preferred.

| RECIPIENT | O+ | O- | A+ | A- | B+ | B- | AB+ | AB- |
|-----------|----|----|----|----|----|----|-----|-----|
| O+        | 1  | 2  | -  | -  | -  | -  | -   | -   |
| O-        | -  | 1  | -  | -  | -  | -  | -   | -   |
| A+        | 3  | 4  | 1  | 2  | -  | -  | -   | -   |
| A-        | -  | 2  | -  | 1  | -  | -  | -   | -   |
| B+        | 3  | 4  | -  | -  | 1  | 2  | -   | -   |
| B-        | -  | 2  | -  | -  | -  | 1  | -   | -   |
| AB+       | 7  | 8  | 3  | 4  | 5  | 6  | 1   | 2   |
| AB-       | -  | 4  | -  | 2  | -  | 3  | -   | 1   |

**Compatibility** for red cells: rows are patients, columns are donor groups, numbers are preference (1 = best).

LifeGrid never suggests a unit or donor outside this table. Exact matches always come first, and O- red cells come last for anyone who isn't O-, so they are kept for those who need them. Switch between red cells, platelets and plasma at the top.

# Audit log

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**Audit log**

sadia.haque@lifegrid.test  
Administrator

Sign out

Demo build. All sites, units, donors and requests are synthetic.

## Audit log

Hash-chained record of every sensitive action. Verification recomputes the whole chain.

 Chain intact. 31 events verified.

Search

| #  | WHEN         | ACTION          | ENTITY                                             | ACTOR                         | HASH             |
|----|--------------|-----------------|----------------------------------------------------|-------------------------------|------------------|
| 31 | Oct 3, 18:29 | setting.change  | setting solver_time_limit_s                        | sadia.haque@lifegrid.test     | e720b2435650a683 |
| 30 | Oct 3, 18:29 | auth.sign_in    | user a612c6fb-6db4-41c7-8b49-b3652977aa49          | sadia.haque@lifegrid.test     | 0054d076b3e9d64a |
| 29 | Oct 3, 18:29 | auth.sign_in    | user 853056b7-dcea-48f5-8abc-1dc8bc43f9df          | imran.chowdhury@lifegrid.test | 379a6592295f40cc |
| 28 | Oct 3, 18:28 | donor.register  | donor d13f1b96-6874-43a8-b905-31f79896bfdc         | user                          | 90d196675e2c4c75 |
| 27 | Oct 3, 18:28 | auth.sign_in    | user b3bd7fcd-de2e-463f-a9e2-3cb79c3a2f3a          | user                          | b05e6fc64d77337d |
| 26 | Oct 3, 18:28 | user.create     | user b3bd7fcd-de2e-463f-a9e2-3cb79c3a2f3a          | user                          | 1ec00a049d9261d7 |
| 25 | Oct 3, 18:28 | match.respond   | donor_match 0674b368-08ad-431c-8a26-1034df3ac3cc   | user                          | 2efb0794c2524ead |
| 24 | Oct 3, 18:28 | auth.sign_in    | user 6dd57e9b-ced3-48a4-8497-d87f3d2c0589          | user                          | 4df0c7c4d841f57f |
| 23 | Oct 3, 18:28 | unit.transition | blood_unit d7273ce7-39c7-4f76-a758-d41b63f619db    | system                        | 7fcf11f5e2282cf1 |
| 22 | Oct 3, 18:28 | unit.transition | blood_unit b3bae46a-a59f-4bdf-8c66-05423ee7deb5    | system                        | cdef5ae3e6865172 |
| 21 | Oct 3, 18:28 | unit.transition | blood_unit 7ba14350-8ce1-470f-a3e4-07af6073a03d    | system                        | 662270811de4a13a |
| 20 | Oct 3, 18:28 | unit.transition | blood_unit 6ea37509-e8bb-4edf-aa5a-30ed1a1b9678    | system                        | 2cc225aeb0a9706e |
| 19 | Oct 3, 18:28 | request.confirm | blood_request 46640153-74b9-414f-963c-16bd242471fb | tanvir.rahman@lifegrid.test   | a57ebf5bfa1fce50 |
| 18 | Oct 3, 18:28 | auth.sign_in    | user 3916a1b0-9afb-45d0-ae09-433baa150023          | tanvir.rahman@lifegrid.test   | d5aa92f9c57eb509 |

## Audit log after Verify chain.

Every sensitive action is recorded: sign-ins and failed sign-ins, role changes, unit changes, transfer decisions, request confirmations, phone sharing, data export and deletion, settings changes. Each event is chained to the one before by a cryptographic hash. **Verify chain** recomputes the whole chain; if any stored event had been edited, LifeGrid names the first one that doesn't match. Auditors can search and verify but change nothing.

# Alerts and reports (preview)

These two screens are part of the next build phase. In this version they show demonstration data only; with the live system they are empty.

## Alerts

The screenshot displays the 'Alerts' section of the LifeGrid application. The left sidebar contains navigation links: Network overview, Site stock, Forecasts, Transfer plan, Transfers, Requests, Alerts (selected, with 16 items), and Reports. The main content area is titled 'Alerts' and includes a subtitle: 'Expiry, stockout, temperature and unfilled-request alerts in your scope.' Below this, there are filters for 'Open' (16), 'Acknowledged' (0), and 'Resolved' (0). The alert list contains the following items:

| Icon | Severity | Category           | Alert Text                                                         | Location                     | Time Ago    | Actions              |
|------|----------|--------------------|--------------------------------------------------------------------|------------------------------|-------------|----------------------|
| 🔥    | Critical | Temperature        | Red cells: storage reached 9.4 °C, 6 units quarantined             | Badda Community Hospital     | 1 hour ago  | Acknowledge, Resolve |
| ⏰    | Critical | Expiry             | Red cells: 10 units expire within 3 days                           | Northern Blood Bank          | 1 hour ago  | Acknowledge, Resolve |
| ⏰    | Warning  | Expiry             | Red cells: 4 units expire within 3 days                            | Motijheel City Hospital      | 1 hour ago  | Acknowledge, Resolve |
| ⏰    | Critical | Expiry             | Red cells: 8 units expire within 3 days                            | Dhanmondi Teaching Hospital  | 2 hours ago | Acknowledge, Resolve |
| ⏰    | Warning  | Expiry             | Red cells: 4 units expire within 3 days                            | Keraniganj District Hospital | 2 hours ago | Acknowledge, Resolve |
| 📉    | Critical | Projected stockout | Red cells O-: forecast demand exceeds stock within 2 days, 3 short | Keraniganj District Hospital | 2 hours ago | Acknowledge, Resolve |
| ⏰    | Warning  | Expiry             | Red cells: 6 units expire within 3 days                            | Mirpur Medical Centre        | 3 hours ago | Acknowledge, Resolve |

The bottom of the sidebar shows the user 'nabila.karim@lifegrid.test' as the 'Blood bank manager' with a 'Sign out' button. A footer note states: 'Demo build. All sites, units, donors and requests are synthetic.'

**Alerts** (demonstration data): expiry, projected stockout, temperature and unfilled-request alerts.

Alerts can be acknowledged (recording who and when) and resolved. Hospital leads see their own sites; coordinators see donor-related alerts.

# Reports

LifeGrid

Regional blood network

Network overview

Site stock

Forecasts

Transfer plan

Transfers

Requests

Alerts

Reports

nabila.karim@lifegrid.test

Blood bank manager

Sign out

Demo build. All sites, units, donors and requests are synthetic.

Reports

Network totals that match the dashboards. Export any report as CSV.

Wastage

Stock

Transfers

Requests

Expired

Discarded

| SITE   | EXPIRED | DISCARDED | ISSUED | WASTAGE % |
|--------|---------|-----------|--------|-----------|
| BB-CEN | 9       | 0         | 11     | 2.8       |
| BB-NTH | 9       | 0         | 7      | 3.2       |
| H-MIR  | 7       | 0         | 9      | 3.9       |
| H-UTT  | 1       | 0         | 3      | 0.8       |
| H-GUL  | 5       | 0         | 3      | 3.3       |
| H-DHN  | 4       | 0         | 9      | 1.6       |
| H-MOT  | 4       | 0         | 6      | 3.4       |
| H-BAD  | 1       | 0         | 1      | 1.3       |
| H-MOH  | 1       | 0         | 0      | 2.2       |
| H-JAT  | 3       | 0         | 5      | 2.1       |
| H-KER  | 1       | 0         | 6      | 1.3       |
| H-SAV  | 3       | 0         | 3      | 3.1       |
| H-TON  | 1       | 0         | 2      | 1.2       |
| H-NAR  | 1       | 0         | 2      | 1.3       |

| SITE   | EXPIRED | DISCARDED | ISSUED | WASTAGE % |
|--------|---------|-----------|--------|-----------|
| BB-CEN | 9       | 0         | 11     | 2.8       |
| BB-NTH | 9       | 0         | 7      | 3.2       |
| H-MIR  | 7       | 0         | 9      | 3.9       |
| H-UTT  | 1       | 0         | 3      | 0.8       |
| H-GUL  | 5       | 0         | 3      | 3.3       |
| H-DHN  | 4       | 0         | 9      | 1.6       |
| H-MOT  | 4       | 0         | 6      | 3.4       |
| H-BAD  | 1       | 0         | 1      | 1.3       |
| H-MOH  | 1       | 0         | 0      | 2.2       |
| H-JAT  | 3       | 0         | 5      | 2.1       |
| H-KER  | 1       | 0         | 6      | 1.3       |
| H-SAV  | 3       | 0         | 3      | 3.1       |
| H-TON  | 1       | 0         | 2      | 1.2       |
| H-NAR  | 1       | 0         | 2      | 1.3       |

**Reports** (demonstration data): wastage, stock, transfers and requests, each exportable as CSV.

# Reference

## Unit statuses

| STATUS      | MEANING                                       | CAN BECOME                                                  |
|-------------|-----------------------------------------------|-------------------------------------------------------------|
| Available   | On the shelf, usable                          | Reserved, in transit, quarantined, expired                  |
| Reserved    | Held for a confirmed request                  | Issued, available (timeout or cancel), quarantined, expired |
| In transit  | Dispatched on an approved transfer            | Available at the destination, quarantined, expired          |
| Quarantined | Held after a temperature excursion or concern | Available (cleared), discarded                              |
| Issued      | Given to a patient                            | (final)                                                     |
| Expired     | Past its expiry time                          | Discarded                                                   |
| Discarded   | Disposed of                                   | (final)                                                     |

## Donor eligibility (defaults)

| RULE             | DEFAULT                                                 |
|------------------|---------------------------------------------------------|
| Age              | 18 to 60                                                |
| Weight           | At least 50 kg (self-reported)                          |
| After a donation | 120 days before the next                                |
| Invitations      | At most 2 per 30 days, unless the donor raises it       |
| Deferral         | Not invited until the eligible-again date               |
| Quiet hours      | No invitations, except emergencies the donor opted into |

## Shelf life and transfers (defaults)

| COMPONENT | SHELF LIFE | MIN. LIFE TO TRANSFER | EXPIRY ALERT |
|-----------|------------|-----------------------|--------------|
| Red cells | 42 days    | 5 days                | 3 days       |
| Platelets | 5 days     | 1 day                 | 1 day        |
| Plasma    | 365 days   | 30 days               | 14 days      |

## Red-cell compatibility

| PATIENT | ACCEPTABLE DONORS, BEST FIRST    |
|---------|----------------------------------|
| O-      | O-                               |
| O+      | O+, O-                           |
| A-      | A-, O-                           |
| A+      | A+, A-, O+, O-                   |
| B-      | B-, O-                           |
| B+      | B+, B-, O+, O-                   |
| AB-     | AB-, A-, B-, O-                  |
| AB+     | AB+, AB-, A+, A-, B+, B-, O+, O- |

## Troubleshooting

| PROBLEM                               | WHAT TO DO                                                                                                                                     |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| No code arrives                       | Check the number includes the country code. Wait a minute; you can request 3 codes an hour. In the demo build the code is shown on screen.     |
| “Too many attempts”                   | The number is locked for 15 minutes after 5 wrong codes.                                                                                       |
| Signed out unexpectedly               | Sessions end after 7 days, when an administrator disables the account, or if an old session token is reused (a safety measure). Sign in again. |
| Another person’s screen appears       | One browser holds one sign-in. Use a separate browser or private window for each person.                                                       |
| “You do not have access to this site” | Your role or site scope doesn’t include it. Ask an administrator.                                                                              |
| Can’t start an appeal                 | An appeal for that group and hospital is already running; follow it on Requests.                                                               |
| Donor not invited                     | Check availability, quiet hours, waiting period, deferral, age, monthly invitation limit and distance from the hospital.                       |
| A screen shows “Something went wrong” | Select <b>Try again</b> . If it persists, the service may be down; your data is safe.                                                          |

## Glossary

| TERM       | MEANING                                                                     |
|------------|-----------------------------------------------------------------------------|
| FEFO       | First-expired, first-out: the unit closest to expiry is used or sent first. |
| Appeal     | A request to donors started by staff when a group runs low at a hospital.   |
| Wave       | One round of invitations. Later waves search a wider area.                  |
| Deferral   | A coordinator's decision that a donor can't donate until a date.            |
| Quarantine | Holding units out of use until they are reviewed.                           |
| Component  | Red cells, platelets or plasma.                                             |
| Rh         | The + or – part of a blood group.                                           |